

Name: _____

PHYS 0525 Analog and Digital Electronics, Spring 2026

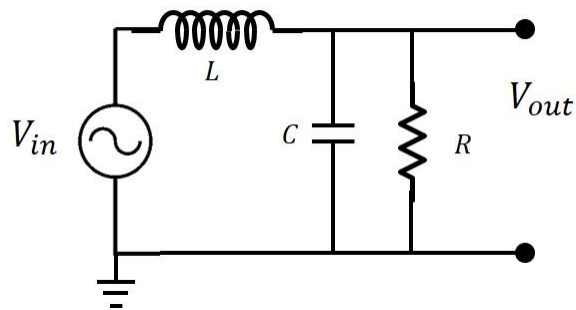
Midterm Exam

Mar. 17, 2026, 11:00am-12:50pm

- Please answer the following questions to complete your midterm exam
- Write all of your solutions on the exam page. If you need extra space, please use the back of the page that the question is written on.
- You may use 1 page of note that you have compiled yourself
- You may use a calculator that does not have an internet connection
- No phones, tablets, laptops, smart watches, or any other devices that are wifi enabled.
- Please box or indicate your final answer to each question
- Please show all of the work necessary to arrive at your final answer
- Please do not begin until instructed to do so.
- Please write your name on the top of the exam

1) (a) Compute V_{out} as a function of V_{in} and the parameters R, C, L, ω for the AC circuit drawn below. Assume the input oscillates sinusoidally at angular frequency ω . You may leave your answer as a complex amplitude.

(b) Describe V_{out} in the limits $\omega \rightarrow 0$ and $\omega \rightarrow \infty$.



$$\left. \begin{array}{l} Z_L = i\omega L \\ Z_C = \frac{1}{i\omega C} \\ Z_R = R \end{array} \right\} Z_C \parallel Z_R = \left(\frac{1}{R} + i\omega C \right)^{-1} = \frac{R}{1 + i\omega RC}$$

$$V_{out} = \left(\frac{Z_C \parallel Z_R}{Z_L + Z_C \parallel Z_R} \right) V_{in} \Rightarrow \frac{\frac{R}{1 + i\omega RC}}{i\omega L + \frac{R}{1 + i\omega RC}} \cdot V_{in} \Rightarrow \frac{V_{out}}{V_{in}} = \frac{\frac{R}{1 + i\omega RC} \cdot 1 + i\omega RC}{i\omega L + \frac{R}{1 + i\omega RC} \cdot 1 + i\omega RC} \Rightarrow \frac{R}{R + i\omega L - \omega^2 LRC}$$

$$\text{SO, } V_{out} = \frac{R}{R + i\omega L - \omega^2 LRC} \cdot V_{in}$$

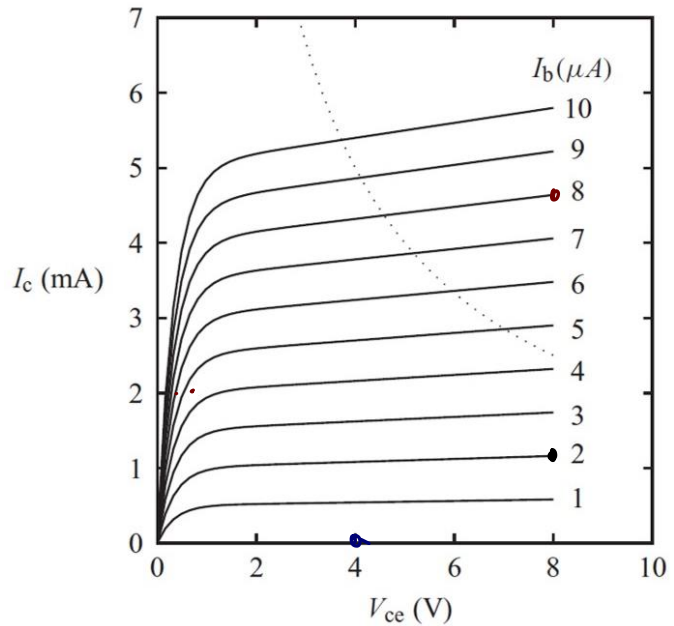
• as $\omega \rightarrow 0$: $Z_L \rightarrow 0$: circuit behaves like a wire

• as $\omega \rightarrow \infty$: $Z_C \rightarrow \infty$: short circuits to ground

2) An *npn* bipolar junction transistor has the I-V curves shown to the right. Use values you extract from these curves to answer the following questions.

(a) Estimate the β value for this transistor if the bias current is approximately $4 \mu\text{A}$?

$$\beta = \frac{I_c}{I_B} = \frac{4 \text{ mA}}{8 \mu\text{A}} = 500$$



(b) If the transistor is in a circuit where the base current is measured to be $8 \mu\text{A}$ and the collector current is measured to be 2 mA , is the transistor operating in active or saturation mode?

in the active region:

$$I = \beta I_B$$

$$\beta \gg 1$$

and independent of V_{ce}

$$2 \text{ E-}3 \text{ A} = 500 \cdot 8 \text{ E-}6 \text{ A}?$$

$$\hookrightarrow 500 \cdot 8 \text{ E-}6 = 4 \text{ E-}3 \neq 2 \text{ E-}3 \text{ A}$$

$I \neq \beta I_B \rightarrow$ saturation mode

(c) If the transistor is in a different circuit where the base current is measured to be $2 \mu\text{A}$ and V_{ce} is measured to be 4 V , is the transistor operating in active or saturation mode? Approximately how much power is dissipated by the transistor in this case?

$$500 \cdot 2 \text{ E-}6 = 1 \text{ E-}3 \text{ A} = 1 \text{ mA}$$

$$\text{at } V_{ce} = 4 \text{ V}$$

$$I_B = 2 \mu\text{A} \Rightarrow \text{active mode}$$

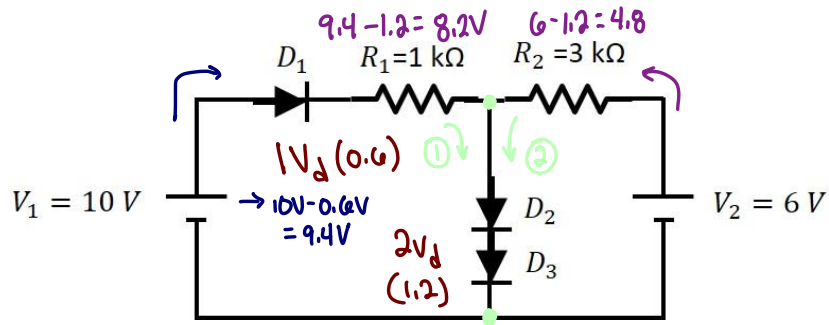
$$I = 1 \text{ mA}$$

$$\text{Power} = VI = V_{ce} I_c + V_{BE} I_B$$

$$* V_{BE} \Rightarrow \text{constant: assume } 0.6 \text{ V}$$

$$\text{So, } P = 4 \text{ V} \cdot 1 \text{ mA} + 0.6 \text{ V} \cdot 2 \mu\text{A} = 0.004 \text{ W} \\ = 4 \text{ mW}$$

3) Calculate the current through each of the diodes, D_1 , D_2 , and D_3 , in the circuit drawn below. Assume the diodes are typical silicon diodes with a diode drop of 0.6 V each.



$$I_{R_1} = \frac{8.2V}{1k\Omega} = 8.2mA$$

so, $I_{D_1} = 8.2mA$

$$I_{R_2} = \frac{4.8V}{3k\Omega} = 1.6mA$$

adding ① + ② $\Rightarrow 8.2mA + 1.6mA = 9.8mA$

and $I_{D_2} = I_{D_3} = 9.8mA$

4) Suppose you have a long coaxial cable of unknown characteristic impedance. You do the following experiment to try to measure its impedance. You terminate one end of the cable with a $200\ \Omega$ resistor. You then send a short pulse into the other end of the cable with an amplitude of 1 V . You measure the reflected pulse to have an amplitude of 0.2 V . What is the impedance of the cable?

$$\frac{\tilde{V}_R}{\tilde{V}_0} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

↑
amplitude

$$* Z_L = 200\ \Omega$$

$$\text{solve for } Z_0: \quad 0.2\text{ V} = \frac{200\ \Omega - Z_0}{200\ \Omega + Z_0} \Rightarrow Z_0 = 125\ \Omega$$

$$\hookrightarrow \text{guess and check} \Rightarrow 0.2\text{ V} = \frac{200 - 125}{325} = \frac{75}{325} = \frac{2}{10} = \frac{1}{5} = 0.2\text{ V} \checkmark$$